

Test and Reliability Exercise Sheet 4

Exercise 1

Assume a fault list $F = \{f_0, \dots, f_{10}\}$ and the following test set, which was generated in the given order, i.e. t_0 was generated first and t_7 was generated last. For each test t_i all faults are given which are detected by this test.

$t_0 = \{f_0\}$	f_0
$t_1 = \{f_0, f_1, f_2\}$ ✓	f_0 , f_1 , f_2
$t_2 = \{f_2, f_3\}$ ✓	f_2 , f_3
$t_3 = \{f_4, f_7\}$ ✓	f_4 , f_7
$t_4 = \{f_5, f_6\}$ ✓	f_5 , f_6
$t_5 = \{f_4, f_7, f_8\}$	f_4 , f_7 , f_8
$t_6 = \{f_9\}$	f_9
$t_7 = \{f_8, f_9, f_{10}\}$ ✓	f_8 , f_9 , f_{10}

→ t_1, t_2, t_3, t_4, t_7

Perform the *Forward-Looking Reverse Order Fault Simulation* for the given test set. Identify for each fault f_j the test t_i which first detects f_j . Then, simulate the test set in reverse order and skip each test t_i which is not the first test to detect any yet undetected fault.

Exercise 2

Gate	AND	NAND	OR	NOR	XOR	XNOR
cont.	0	0	1	1	—	—
noncont.	1	1	0	0	—	—

- State for the following set of 2-input gates the controlling values as well as the non-controlling ones: $\{AND, NAND, OR, NOR, XOR, XNOR\}$
- Reconsider the example circuit for static test pattern relaxation from the lecture as given in Fig. 1. In the lecture we obtained the test pattern $t_g = 1X11$ (by utilizing a greedy approach) and $t_o = 11XX$ (as optimal solution in number of unspecified inputs) for the stuck-at-fault $s@0$ at the input A (see Fig. 1). Find another stuck-at-fault in the circuit for which a (relaxed) test pattern can be statically compacted with t_g , but not with t_o .

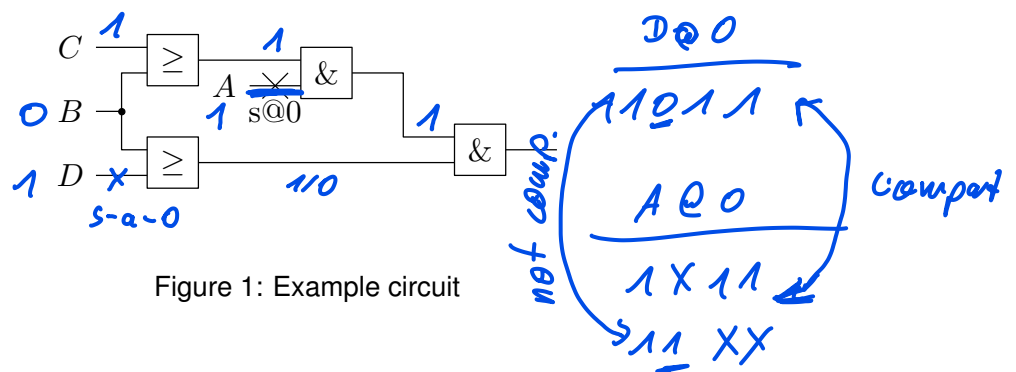


Figure 1: Example circuit

Exercise 3

- Perform a cone-of-influence (COI) reduction on the circuit in Fig. 2 with a stuck at 0 fault at line x.
- Build a miter circuit with the relevant part of the circuit.

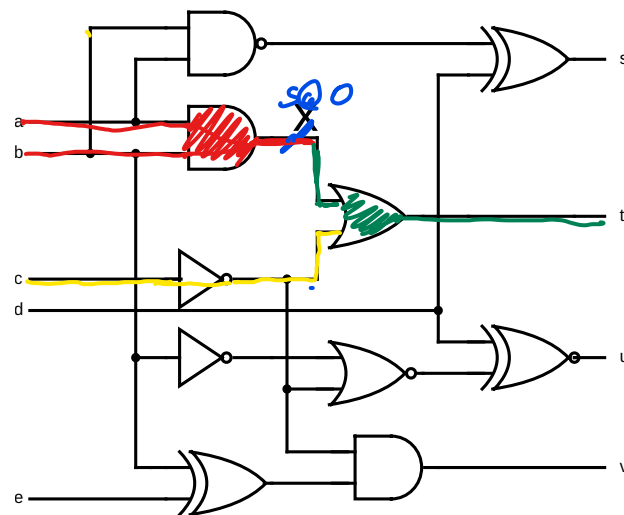
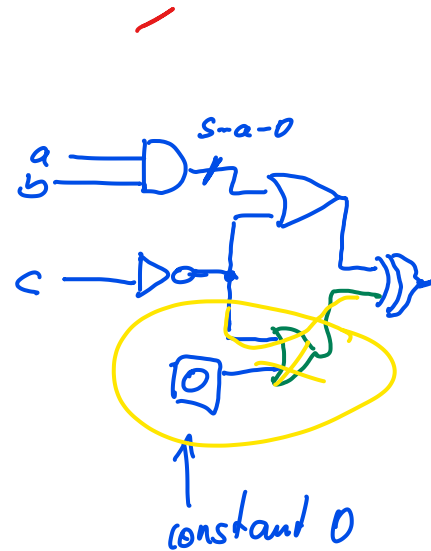


Figure 2: Cone-of-Influence reduction circuit



Exercise 4

- Perform a Tseitin transformation on the circuit from the lecture as given in Fig. 3.
- Perform the SAT-Solving algorithm (chapter 4, slides 35ff) from the lecture on the SAT-instance you obtained from part a) including the unit clause “set BD to 1”. Whenever you reach a conflict, go back to the last decision you have made. Additionally, use the following decision heuristic:
 - When you have to make a decision, choose variables close to the inputs of the original circuit first. If there are still multiple possibilities, choose an arbitrary one.
 - When you have chosen a variable, assign the logical value 0 first to this variable.
- Compare the circuit with your solution of exercise 3b.

Deadline: Discussion on Thursday, June 25th, 2026

$$① d \equiv (a \wedge b)$$

$$② e \equiv \neg c$$

$$③ f \equiv (d \vee e)$$

$$④ d' \equiv 0$$

$$⑤ e' \equiv \neg c$$

$$f' \equiv (d' \vee e')$$

$$BD \equiv (f \oplus f')$$

$$BD \equiv 1$$

$$\begin{matrix} d'=0 \\ BD=1 \end{matrix}$$

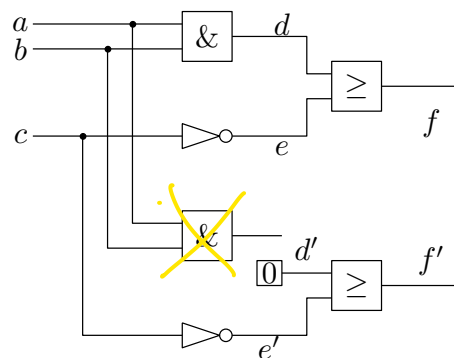
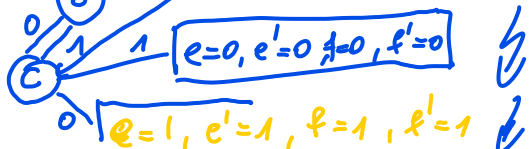
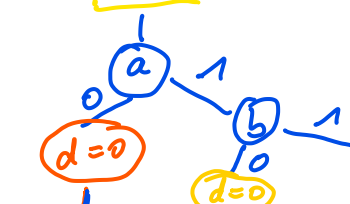


Figure 3: Miter circuit

$$\begin{aligned} ① d &\rightarrow (a \wedge b) \wedge (a \wedge b) \rightarrow d \\ &\equiv ((a \wedge b) \vee \neg d) \wedge (d \vee \neg(a \wedge b)) \\ &\equiv (a \vee \neg d) \wedge (b \vee \neg d) \wedge (d \vee \neg a \vee \neg b) \end{aligned}$$

$$② (e \vee c) \wedge (\neg e \vee \neg c)$$

$$\begin{aligned} ③ f &\rightarrow (d \vee e) \wedge (d \vee e) \rightarrow f \\ &\equiv (d \vee e \vee \neg f) \wedge (f \vee \neg(d \vee e)) \\ &\equiv (d \vee e \vee \neg f) \wedge (f \vee \neg d) \wedge (f \vee \neg e) \end{aligned}$$

$$④ \neg d' \quad ⑤ (e' \vee c) \wedge (\neg e' \vee \neg c)$$

$$\begin{aligned} SAT: & (a \vee \neg d) \wedge (b \vee \neg d) \wedge (\neg a \vee \neg b \vee d) \wedge (c \vee e) \wedge (\neg c \vee \neg e) \\ & \wedge (\neg d \vee f) \wedge (\neg e \vee f) \wedge (d \vee e \vee \neg f) \wedge (c \vee e') \wedge (\neg c \vee \neg e') \\ & \wedge (\neg d' \vee f') \wedge (\neg e' \vee f') \wedge (d' \vee e' \vee \neg f') \wedge \\ & (\neg f \vee f' \vee BD) \wedge (f \vee \neg f' \vee BD) \wedge (f \vee f' \vee \neg BD) \wedge (\neg f \vee \neg f') \wedge BD \\ & \wedge (\neg d') \wedge (BD) \end{aligned}$$

$$d=1, f=1, f'=0, e'=0, c=1, e=0$$